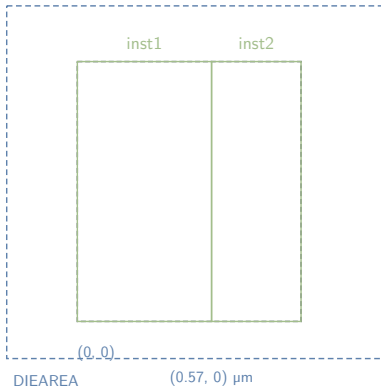


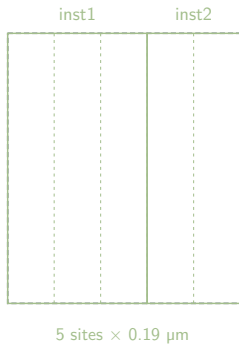
## DEF: place two instances



```
# design.def
DESIGN design ;
UNITS DISTANCE MICRONS 2000 ;
DIEAREA ( -600 -400 ) ( 2600 3400 ) ;

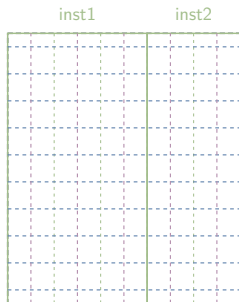
COMPONENTS 2 ;
- inst1 NAND2_X1 + PLACED ( 0 0 ) N ;
- inst2 INV_X1 + PLACED ( 1140 0 ) N ;
END COMPONENTS
```

## DEF: rows are made of sites



```
# design.def
ROW ROW_0
  FreePDK45_38x28_10R_NP_162NW_340
  0 0 N ← origin / orientation
  D0 5 BY 1 ← count / step
  STEP 380 0 ;
```

# DEF: tracks guide routing

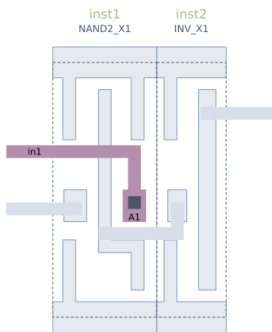


5 sites × 0.19 μm

```
# design.def
TRACKS Y 140 ← horizontal M1
DO 10 STEP 280
LAYER metal1 ;

TRACKS X 190 ← vertical M2
DO 5 STEP 380
LAYER metal2 ;
```

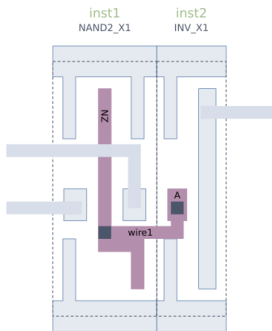
# DEF: ports at the design boundary



Highlighted net: in1

```
# design.def
PINS 3 ;
- in1 + NET in1 ← block terminal
+ DIRECTION INPUT + USE SIGNAL
+ LAYER metal2 ( -70 -70 ) ( 70 70 ) ← shape / position
+ PLACED ( -440 1820 ) N ;
# in2 and out omitted
END PINS
```

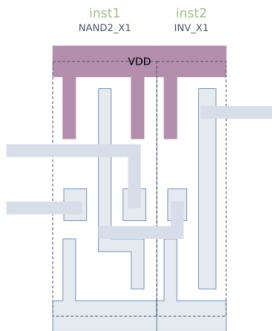
# DEF: connectivity and metal routes



Highlighted net: wire1

```
# design.def
NETS 4 ;
# Other nets omitted
- wire1 ( inst1 ZN ) ( inst2 A ) ← two instance terminals
+ ROUTED metal2
  ( 570 920 ) ( 1365 920 )
  ( 1365 1190 )
  NEW metal2 ( 570 920 ) via1_5 ← route + via
  NEW metal2 ( 1365 1190 ) via1_5 ;
END NETS
```

# DEF: the power connections



Highlighted net: VDD

```
# design.def
SPECIALNETS 2 ;
- VDD ( inst1 VDD ) ( inst2 VDD ) ← power connectivity
+ USE POWER
+ ROUTED metal1 340 ← explicit wire width
  ( 0 2800 ) ( 1900 2800 ) ;
- VSS ( inst1 VSS ) ( inst2 VSS )
+ USE GROUND
+ ROUTED metal1 340
  ( 0 0 ) ( 1900 0 ) ;
END SPECIALNETS
```